• S (Entropy) $= K_B \log$ (total number of micro states).

Total number of microstates

$$= \frac{\text{Phase space satisfying constraint}}{h^{3N}}$$

Constraint $\rightarrow$ (1) Total volume V
(2) Total energy E } Microcanonical ensemble!

$$\text{Phase space} = \int dx_1 \ldots dx_{3N}\, dp_1 \ldots dp_{3N}.$$

I used volume constraint.

$$= V^N \int \underbrace{dp_1 \ldots dp_{3N}}_{} \quad \text{—— (I)}$$

I have to still use momentum constraint to evaluate

$$\frac{1}{2m}\left(p_1^2 + \ldots + p_{3N}^2\right) = E \longrightarrow (A)$$

This is eqn of a sphere of 3N dimension

Note that, ~~any~~ at any moment of time, the SYSTEM is described by a point in the phase space and that point must lie on the sphere satisfying (A)

From (I) we have to calculate

$$\int dp_1 \cdots dp_{3N} = \text{Surface area of the sphere}$$

A. Now you can calculate surface area by taking derivative of volume ie. $\left.\frac{dV}{dr}\right|_{r=\sqrt{2mE}}$

where $V = \frac{\pi^{n/2}}{(\frac{n}{2})!\,\Gamma(\frac{n}{2})} r^n$

Class notes

This is the procedure we had taken in the class

we wrote that as $d\Omega$ & accordingly

$$S = k_B \ln(d\Omega)$$

* Do not worry about "d" part. We care about # of microstates!

B. You can calculate surface area by using

$$A = \frac{\pi^{n/2}}{\frac{1}{2}\Gamma(\frac{n}{2})} r^{n-1}$$ (Formula given)

Current HW #9

Angular part

Then $\Omega = \left(A\big|_{r=\sqrt{2mE}} \cdot V^N \right) / h^{3N}$.

$$S = k_B \log \Omega$$

Both A & B same !!

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